

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456496.6967 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.047
Transit depth (Rp/Rs)^2	2.53 +/- 1.49 [%]
Orbital Inclination (inc)	85.84 +/- 2.91
Airmass coefficient 1 (a1)	4.46 +/- 2.34
Airmass coefficient 2 (a2)	-1.16 +/- 0.45
Scatter in the residuals of the lightcurve fit is	48.87 %
Best Comparison Star	#1 - [266, 181]
Optimal Aperture	1.82
Optimal Annulus	5.5
Transit Duration (day)	0.037 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.047 vs 0.1326 (deviation 0.28σ)
Duration fit vs published	0.037 d vs 0.0737 d (deviation 1.14σ)
Mid-time O-C	-7.3 min
Depth SNR	1.7
Residual scatter	48.87 %

Light curve

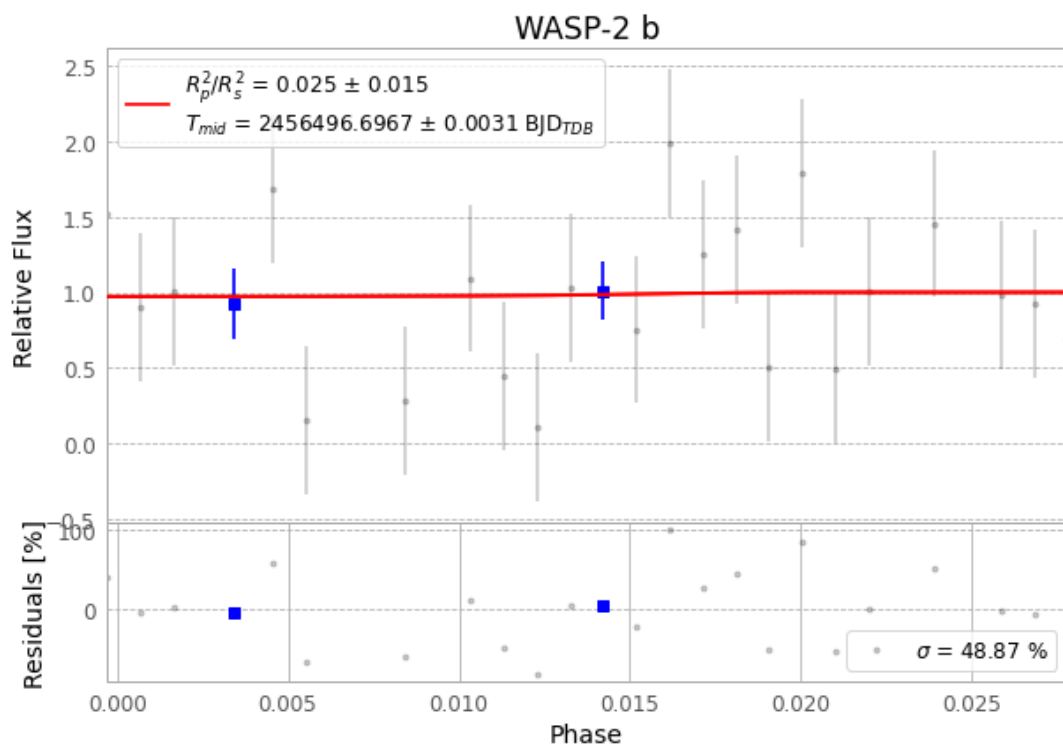


Figure 1 — FinalLightCurve_WASP-2 b_2013-07-22.png (SHA-256 cefec45d8ee0e0ac...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [199, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f53b703e669785c2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

52 FITS frames (34 MB), WASP-2130723034213.FITS → WASP-2130723061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-130723.log (4bb13af8fef3...), FinalLightCurve_WASP-2 b_2013-07-22.png (cefec45d8ee0...), FinalLightCurve_WASP-2 b_2013-07-22.csv (eac451783918...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:36Z.